Mario Michelessa

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Human–Computer Interaction and Human-Centered AI

Research Areas

Human-centered AI; interactive machine learning; visual analytics; image collections; computer vision.

Education

PhD in Computer Science Jan 2023 - Jan 2027
National University of Singapore, School of Computing *Singapore*
Dissertation: Scalable Human-AI Interaction for Image Collections
Advisors: Prof. Brian Y. Lim, Prof. Christophe Hurter

- Designed and evaluated interactive AI systems for exploring, generating, and interpreting large image collections.
- Conducted user studies combining quantitative and qualitative methods to study interaction with AI systems, visual representations, and navigation aids.
- Developed machine-learning methods and theoretical analyses for transformer expressivity, prompt tuning, and patient-relative image classification.

MSc by Research in Statistics and Applied Probability Aug. 2021 – Dec. 2022
National University of Singapore *Singapore*
Thesis: Decision-Focused Models for Influence Maximization in Social Networks
Advisors: Prof. Carol-Anne Hargreaves, Prof. Bogdan Cautis, Prof. Xiaokui Xiao

Combined MSc and Engineering Degree Sep. 2018 – Dec. 2022
CentraleSupélec, Université Paris-Saclay *France*

Publications

Peer-Reviewed Publications

1. Maxime Meyer*, **Mario Michelessa***, Caroline Chaux, and Vincent YF. Tan. *How Many Different Outputs Can a Transformer Generate?* International Conference of Machine Learning (ICML '26 Oral, top 0.8% of submissions). https://arxiv.org/abs/2605.22223
2. **Mario Michelessa**, Jamie Ng, Christophe Hurter, and Brian Y. Lim. 2025. *Varif.ai to Vary and Verify User-Driven Diversity in Scalable Image Generation.* Designing Interactive Systems (DIS '25). https://doi.org/10.1145/3715336.3735847
3. **Mario Michelessa**, Christophe Hurter, Brian Y. Lim, Jamie Ng Suat Ling, Bogdan Cautis, and Carol Anne Hargreaves. 2023. *Visual Explanations of Differentiable Greedy Model Predictions on the Influence Maximization Problem.* Big Data and Cognitive Computing, 7(3), 149. https://doi.org/10.3390/bdcc7030149

Preprints

1. Wencan Zhang*, **Mario Michelessa***, Xuejun Zhao, and Brian Y. Lim. 2026. *SketchXplain: Intuitive Visual Explanations of Image Classifiers with Sketches.* arXiv preprint. https://arxiv.org/abs/2606.17646
2. Maxime Meyer, **Mario Michelessa**, Caroline Chaux, and Vincent YF Tan. 2025. *Memory Limitations of Prompt Tuning in Transformers.* arXiv preprint https://arxiv.org/abs/2509.00421.

Under Review

1. **Mario Michelessa**, Jamie Ng, Christophe Hurter, and Brian Y. Lim. 2025. *ReAxis: Interactive Refinement of Axes for Exploratory Visual Analysis of Image Collections.* IEEE Transactions on Visualization and Computer Graphics.
2. **Mario Michelessa**, Floriane Payen, Alexandre Veyrie, Maria-Jesus Lobo-Gunther, Arnaud Prouzeau and Christophe Hurter. *Singing Directions: An Exploratory Study on Musical Navigation and Route Memorization.* Taylor & Francis International Journal of Human-Computer Interaction

*Co-first authors, equal contribution

Research Experience

University of Queensland — Visiting Researcher May 2026 – Jul. 2026
School of Electrical Engineering and Computer Science *Brisbane, Australia*

- Developed patient-relative melanoma-detection models that compare each lesion with other lesions from the same patient, following dermatology's "ugly duckling" principle. Supervised by Prof. Tim Miller and Prof. Maxime Cordeil.

UbiComp Lab, National University of Singapore — Research Intern Sep. 2022 – Dec. 2022
School of Computing *Singapore*

- Developed edge-bundled visualizations for inspecting influence propagation and selected influential nodes in large social networks.
- Investigated how AI could support aircraft-maintenance tasks, supervised by Prof. Brian Y. Lim and Prof. Christophe Hurter.

CNRS@CREATE — Research Intern Feb. 2022 – Aug. 2022
Descartes Program *Singapore*

- Developed learning-based methods for selecting influential users in social networks by making the influence-maximization objective differentiable. Supervised by Prof. Xiaokui Xiao and Prof. Bogdan Cautis.

Awards and Funding

- **AISG–Descartes Joint PhD Scholarship** (Jan. 2023–Jan.2027), one of Singapore's most selective national AI research scholarship programs, under the DesCartes Program with CNRS@CREATE.

Selected Invitations

- **Invited Participant**, Dagstuhl Seminar 24301, "Art, Visual Illusions, and Data Visualization," Schloss Dagstuhl, Germany, Jul. 2024.
- **Invited Participant**, 2nd MERCADO Workshop, IEEE VIS 2026, Vienna, Nov. 2026.

Teaching and Mentoring

Teaching

- **Teaching Assistant**, CS4249 Quantitative Methods of Human-Computer Interaction, National University of Singapore, AY2023/24 and AY2024/25. Led tutorials, developed teaching materials, and graded assignments.
- **Teaching Assistant**, CS3244, Deep Learning, National University of Singapore. Coordinated project assessment across teaching assistants and developed teaching materials.
- **Invited Teaching Assistant**, Information Visualization, MSc in Human–Computer Interaction, ENAC, Assisted students during practical sessions and project development.

Mentoring

- **Undergraduate Research Mentor**, National University of Singapore, Jolyn Loh, 2024. Explainable skin-lesion classification using dermatological concepts and concept-bottleneck models.

Academic Service

- **Reviewer:** IEEE VIS 2026; ACM DIS 2026; ACM CHI 2026; IEEE TVCG 2026; IEEE Pacific Visualization 2024 and 2026.
- **Safety Lead**: Ubicomp Lab, 2023–2026. Coordinated laboratory safety procedures, documentation, and compliance.
- **Server Administrator**: Ubicomp Lab, 2023–2026. Maintained shared research servers, user access, and computing infrastructure.

Invited Talks

- "[I²R Sparks Talk Series] Varif.ai to Vary and Verify User-Driven Diversity in Scalable Image Generation," A*STAR, I²R, Advanced & Sustainable Manufacturing (ASM), Singapore, Aug. 2025.

Industry Experience

Data Scientist Intern Sep. 2020 – Feb. 2021
BNP Paribas Cash Management *Paris, France*
Work: Developed fraud-detection models on large-scale transaction data and interpretable features for fraud analysts.

Technical Expertise

Programming: Python, PyTorch, React/TypeScript, R, SQL, LaTeX.
Machine learning: Deep learning, computer vision, vision–language models, explainable AI;
Quantitative methods:: Bayesian modeling, mixed-effects models, experimental design, statistical inference;
HCI methods: Interactive system design; controlled experiments; interviews and qualitative analysis.